

Fabrication and Characterization of Nanometer-Thick PDMS Films for Hard Disk Drives

Fajer Almanea¹, Xuhao Zhou², Qian Guo², and Lei Li¹

¹Department of Chemical and Petroleum Engineering, University of Pittsburgh, Pittsburgh, PA 15261, USA; lel55@pitt.edu

²Recording Media Operations, Seagate Technology LLC, Fremont, California, USA, xuhao.zhou@seagate.com

Nanometer-thick media lube is critical to hard disk drive (HDD). The industry-standard, e.g., perfluoropolyether (PFPE), belongs to the broader class of per- and polyfluoroalkyl substances (PFAS), which pose significant environmental and human-health concerns due to their persistence, toxicity, and bioaccumulation. Increasing regulations and public scrutiny have accelerated global efforts to reduce PFAS usage, creating long-term regulatory risk for the HDD industry and an urgent need for non-PFAS alternatives. Polydimethylsiloxane (PDMS) is of interest owing to its intrinsically low surface tension, chemical stability, and tunable chemistry. However, its performance as a nanometer-thick film remains underexplored. Here, hydroxyl-terminated PDMS films were deposited on native-oxide Si wafers by dip-coating and characterized by spectroscopic ellipsometry, atomic force microscopy (AFM), and static water/hexadecane contact angle measurements. The roughness of PDMS nanofilm is comparable to bare Si wafer, with no dewetting across the concentration range examined. At ~ 1.3 nm, water and hexadecane contact angles approach those of bulk polytetrafluoroethylene, confirming the low-surface-energy character at thicknesses directly relevant to the HDI. The results suggest that hydroxyl-terminated PDMS could be a promising candidate for HDD lubrication.

Index Terms—Atomic force microscopy, contact angle, dip coating, hard disk drive lubricant, polydimethylsiloxane, thin film, PDMS.

I. INTRODUCTION

The head-disk interface (HDI) in hard disk drives (HDDs) is protected by a nanometer-thick lubricant, i.e., perfluoropolyether (PFPE), whose uniformity, bonded/mobile balance, and low surface energy govern tribological reliability. [1], [2]. PFPE belongs to the broader class of per- and polyfluoroalkyl substances (PFAS), which pose significant environmental and human-health concerns due to their persistence, toxicity, and bioaccumulation. Increasing regulations and public scrutiny have accelerated global efforts to reduce PFAS usage, creating long-term regulatory risk for the HDD industry and an urgent need for non-PFAS alternatives. Polydimethylsiloxane (PDMS) is a promising candidate: its flexible siloxane backbone and outward-facing methyl groups yield a surface tension of ~ 20 mN/m, among the lowest of any non-fluorinated polymer [3]. PDMS is thermally and chemically stable, and hydroxyl-terminated variants can interact strongly with oxide-bearing surfaces. While PDMS is widely studied in bulk and as micrometer-scale coatings [4], its behavior as a nanometer-thick film on smooth substrates in the thickness regime directly relevant to the HDI has received limited attention. Here, we report a repeatable dip-coating protocol for PDMS nanofilms on native-oxide Si using Vertrel XF as the carrier solvent, following the procedure established for PFPE lubricants on HDD media [1]. The PDMS nanofilms were characterized by spectroscopic ellipsometry, atomic force microscopy (AFM), and static contact angle measurements with water and hexadecane. Total and bonded thicknesses were quantified as functions of concentration and dwell time. Smooth films with sub-nanometer roughness were obtained, and at ~ 1.3 nm, the coatings exhibited low-surface-energy behavior, approaching reported bulk polytetrafluoroethylene (PTFE) contact angles [5].

II. EXPERIMENTAL METHODS

Hydroxyl-terminated PDMS ($M_w \approx 2000$ g/mol, 65 cSt) was dissolved in 2,3-dihydrodecafluoropentane (Vertrel XF) at 0.025–5 g/L and dip-coated onto silicon wafers with native oxide at a withdrawal speed of 60 mm/min, with dwell times of 20 s to 3 min. As-coated films contain both bonded and mobile PDMS fractions; the bonded fraction was determined after two sequential Vertrel XF rinse cycles (20 s dwell, 60 mm/min), which remove the mobile fraction [1]. Film thicknesses were measured by spectroscopic ellipsometry (J. A. Woollam) using the native-oxide thickness fit first, and the PDMS layer was modeled with a Cauchy dispersion ($n \approx 1.43$). Surface morphology was measured by AFM in tapping mode; static water (WCA) and hexadecane (HCA) contact angles were recorded using the sessile-drop method.

III. RESULTS AND DISCUSSION

A. Thickness vs. Solution Concentration

Fig. 1 shows total, bonded, and mobile PDMS film thicknesses versus solution concentration at a fixed dwell time of 20 s. Total and bonded thicknesses increase monotonically with concentration over 0.025–5 g/L, consistent with dip-coated polymer films in the drainage-dominated regime [1]. Total thickness rises from 0.14 ± 0.03 nm at 0.025 g/L to ~ 3.7 nm at 5 g/L, spanning the sub-nanometer to few-nanometer range directly relevant to HDD lubrication. The bonded fraction accounts for most of the deposited film at every concentration (e.g., 0.11 ± 0.01 nm of 0.14 nm at 0.025 g/L; ~ 2.7 nm of 3.7 nm at 5 g/L), while the mobile fraction (total minus bonded) remains below ~ 1.1 nm throughout. This partitioning indicates that a substantial portion of the hydroxyl-terminated PDMS anchors to the native-oxide surface, while the remainder forms an overlayer removed by rinsing, consistent with the behavior previously reported for hydroxyl-terminated PFPEs on HDD media [1], [2].

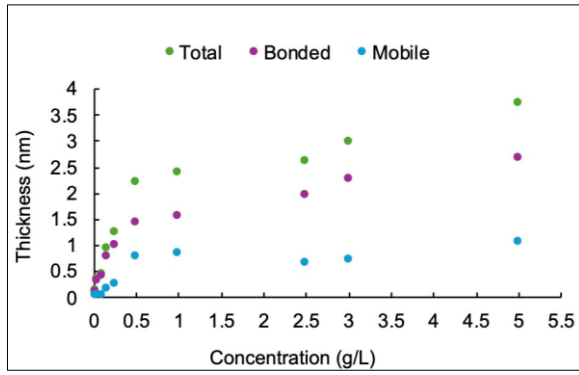


Fig. 1. Total and bonded, and mobile PDMS thickness as a function of solution concentration (dwell time 20 s, withdrawal speed 60 mm/min). Mobile = total – bonded.

B. Surface Morphology and Roughness

Fig. 2 shows AFM images and R_q values for representative PDMS films before and after Vertrel XF washing, with a bare Si included as a reference ($R_q = 0.160$ nm). All films are continuous, with no evidence of dewetting, pinholes, or aggregated droplets. Before washing, R_q ranges from 0.184 to 0.260 nm, marginally above the bare-substrate value. After washing, R_q decreases at every concentration (the largest decrease at 0.25 g/L: 0.209 to 0.145 nm); post-wash values at 0.1 and 0.25 g/L are slightly lower than those of bare Si, suggesting that the bonded PDMS layer forms a conformal coating over the native oxide surface. The absence of dewetting is notable: hydroxyl-terminated PFPEs on amorphous carbon undergo autophobic dewetting above a molecular-weight-dependent critical thickness [6]. Here, no dewetting is observed for hydroxyl-terminated PDMS on native oxide throughout the thickness range studied.

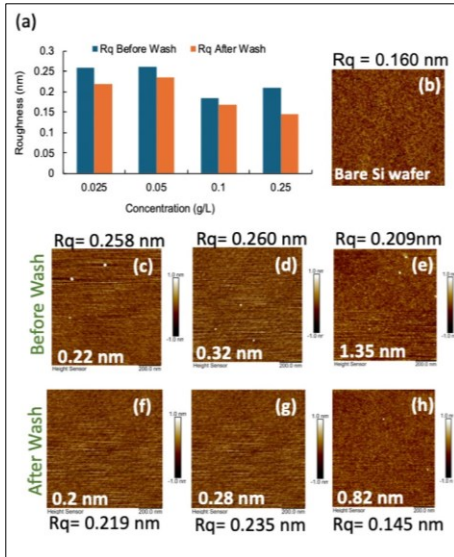


Fig. 2. AFM characterization of PDMS films before and after Vertrel XF washing. (a) R_q bar chart. (b) Bare Si reference ($R_q = 0.160$ nm). (c-e) As coated films at 0.025, 0.05, and 0.25 g/L. (f-h) Corresponding films after washing. Scan size 200×200 nm; z-scale ± 1.0 nm.

C. Wettability and Low Surface Energy

Fig. 3 shows static water (WCA) and hexadecane (HCA) contact angles on as-coated PDMS films at five thicknesses (0.17–1.31 nm, 1 min dwell time). For reference, bare Si

measured in this study exhibits $WCA = 37.2^\circ$ and $HCA = 15.4^\circ$. Both angles increase consistently with PDMS thickness: WCA rises from 52.0° at 0.17 nm to 96.1° at 1.31 nm, and HCA from 12.0° to 49.7° . Even the thinnest film raises the WCA well above bare Si, indicating that sub-monolayer coverage substantially modifies interfacial chemistry. At 1.31 nm, both angles approach the reported (PTFE) contact angles, with $WCA \approx 110^\circ$ and $HCA \approx 47^\circ$ [5]. This behavior is consistent with the well-established low surface-energy character of methyl-terminated silicone surfaces [3]. The approach to PTFE-like wetting at ~ 1 nm confirms acquisition of low surface-energy character at thicknesses directly relevant to the HDI.

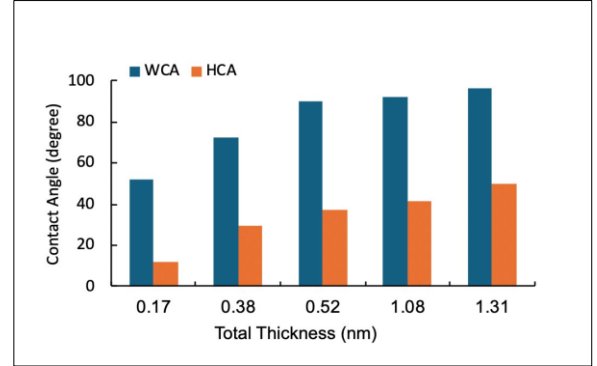


Fig. 3. Static water (WCA) and hexadecane (HCA) contact angles as a function of total PDMS thickness (dwell time 1 min).

IV. CONCLUSION

A repeatable dip-coating process for hydroxyl-terminated PDMS nanofilms on native-oxide Si wafer has been established, producing films from sub-nanometer to several nanometers thick. Solution concentration is the dominant control of total and bonded thickness, while dwell time provides secondary tuning. AFM confirms continuous, smooth films with no dewetting, and post-wash roughness is at or below that of bare Si. Water and hexadecane contact angles increase steadily with thickness, approaching PTFE contact-angle values at ~ 1.3 nm. These results establish PDMS nanofilms as a viable low-surface-energy coating at HDI-relevant thicknesses.

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